

ENCN445: Environmental Fluid Mechanics

Coastal Engineering Tutorials:

Module Four

Problem One: While stationary, a canoer drags the cylindrical handle of their paddle in the ocean. They notice that the force on the paddle seems to change over time. After some research they discover that the force acting on the paddle could be modelled with the Morrison equation. The amplitude of the waves when they were paddling was 0.1 m, the period was 6 s, and the phase speed of the wave was 8 m/s. The paddle was 3 cm in diameter and the canoer was in relatively deep water so the depth of the submerged paddle was negligible compared to the fluid depth (i.e. assume that $\cosh(k(z + h)) = \cosh(kh)$). The density of the ocean at this location was 1025 kg/m^3 .

A Use the Morrison equation to find the maximum drag force and inertial force acting in the x direction on the paddle per unit length. Use a drag coefficient of 1 and an inertial coefficient of 2.

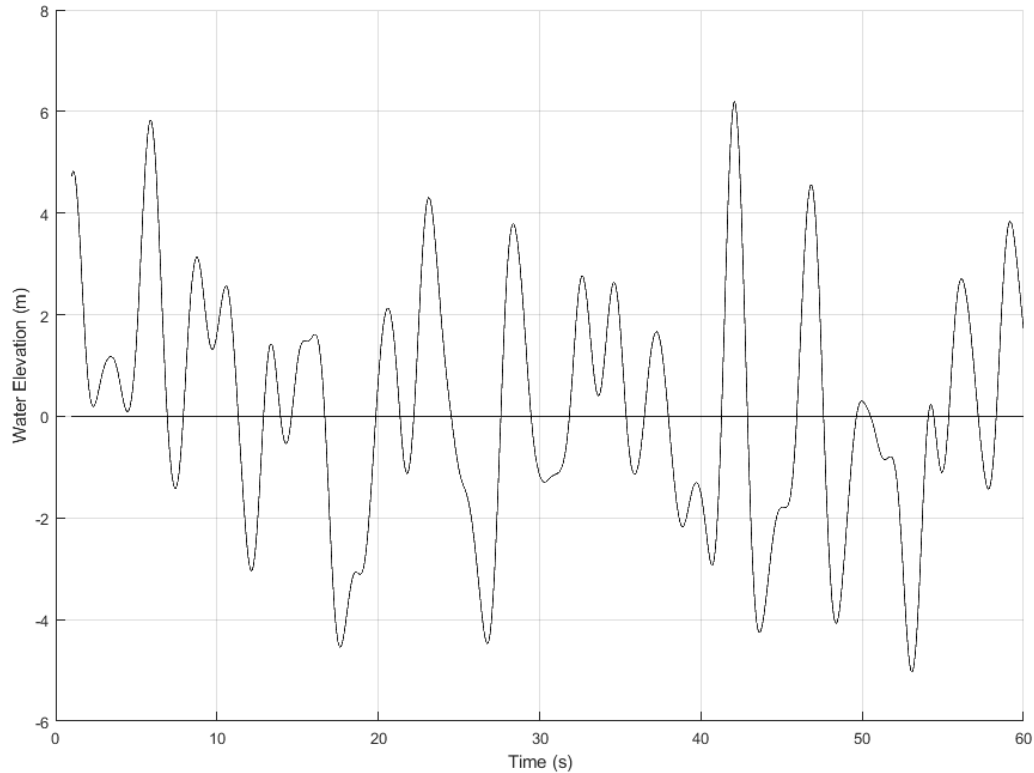
Problem Two: Some young New Zealanders have forgone avocado on toast for several decades and are now looking at buying a house. They are looking at a beachfront property but are concerned with the chance of their front lawn flooding during a storm.

A They decide to calculate how fast the wind would have to blow to induce a wind setup that would flood their lawn. The lawn is 30 cm above the normal high tide line and the continental shelf out from the house is 40 km wide and 50 m deep. The ocean density is 1025 kg/m^3 . Use a friction coefficient of 2×10^{-6} . Calculate the sustained wind speed needed over the continental shelf to flood the front lawn and advise the potential house buyers how big a risk you think this is.

B The potential homeowners recognise that wind setup isn't the only component of a storm surge. The baroclinic storm surge is approximately 1 cm for every hPa drop in air pressure below 1010 hPa. Assume that a storm has a minimum pressure of 990 hPa and calculate the required windspeed for the total storm surge to overtop the front lawn.

Problem Three:

A client has commissioned a wave survey in preparation for an upcoming project. A representative wave record from typical conditions is shown below.



- A** Use an up-crossing analysis to calculate the mean wave height H_z , the mean wave period T_z , and the significant wave height H_s .
- B** The client is interested in extreme events so wants to know the probability of a wave taller than 15 m in this location. Assuming that waves follow a Rayleigh distribution, calculate the probability of a given wave we be at least 15 m tall.
- C** Use the spectral width to test how well this wave follows a Rayleigh distribution.